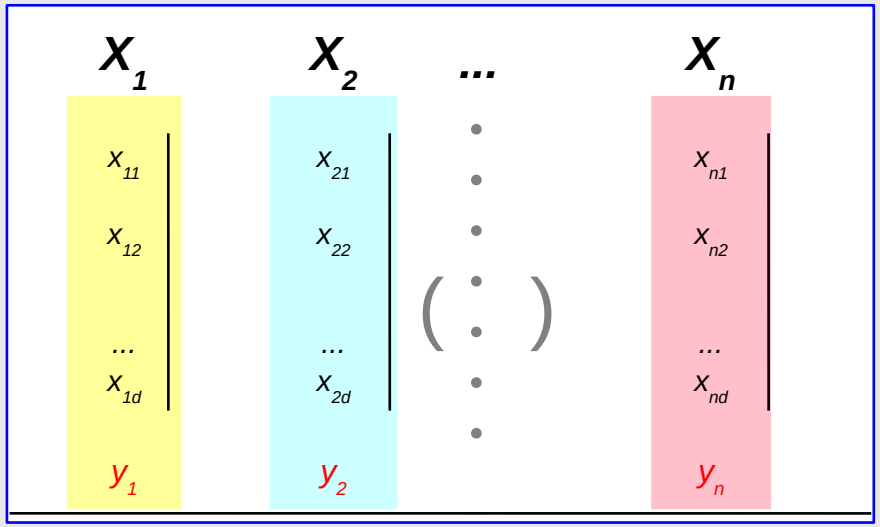
Matrix Notation for Linear Regression and Classification (N- Samples)

N Training Samples



Remember

$$y = \underline{W}^T\underline{X} = \underline{X}^T\underline{W}$$

In General: **N >>> d**

$$\begin{bmatrix} y_1 \\ y_2 \\ y \\ y_n \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1d} \\ x_{21} & x_{22} & \dots & x_{2d} \\ \dots & \dots & \dots & \dots \\ x_{n1} & x_{n2} & \dots & x_{nd} \end{bmatrix} \begin{bmatrix} b \\ w_1 \\ w_2 \\ \dots \\ w_d \end{bmatrix}$$

N × 1 — N × d+1 — d+1 × 1

$$\underline{Y} = \underline{X}\,\underline{W}$$

Where: $\underline{X}_{N\times d+1}$ *is a matrix*

$\underline{Y}_{N\times 1}$ *is labels vector*